

WASP-12b

R-0118

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_251205 · created 2026-07-16T04:47:24+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461014.9247 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.027
Transit depth (Rp/Rs)^2	2.38 +/- 0.84 [%]
Orbital Inclination (inc)	83.75 +/- 2.48
Airmass coefficient 1 (a1)	1.176 +/- 0.015
Airmass coefficient 2 (a2)	-0.121 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	None
Optimal Aperture	3.64
Optimal Annulus	5.04
Transit Duration (day)	0.1119 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.027 vs 0.1178 (deviation 1.34σ)
Duration fit vs published	0.1119 d vs 0.125 d (deviation 4.23σ)
Mid-time O–C	-4.3 min
Depth SNR	5.7
Residual scatter	1.29 %

Light curve

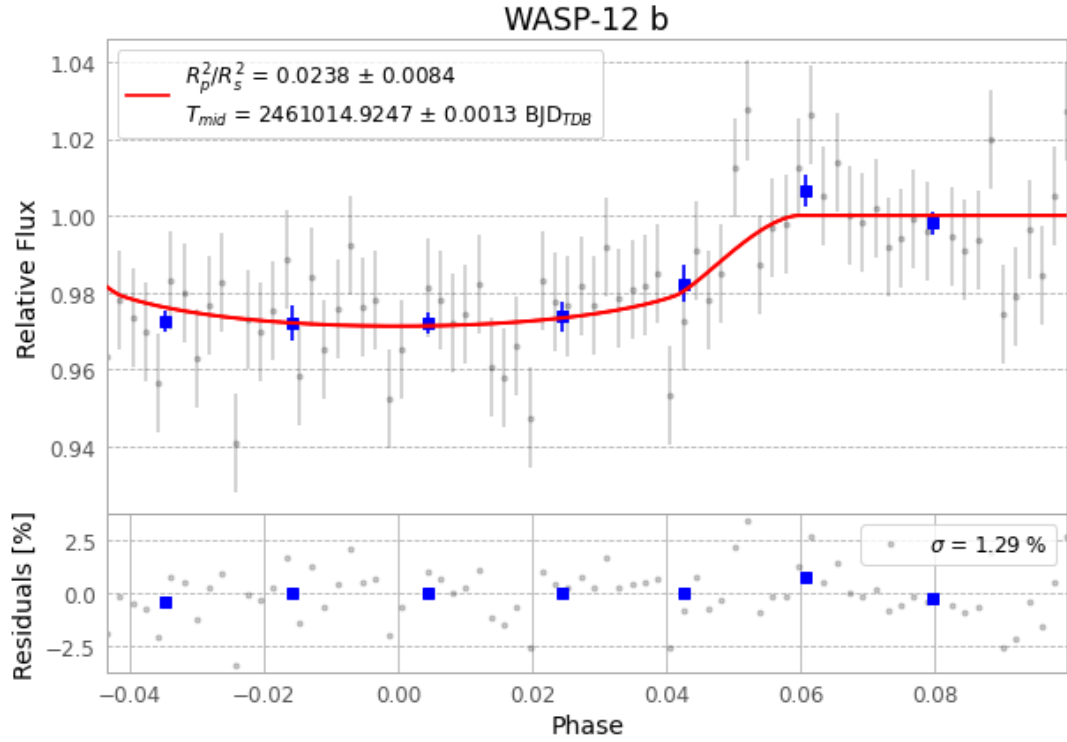


Figure 1 — FinalLightCurve_WASP-12 b_2025-12-05.png (SHA-256 a2bf50c5d1ede2cd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [377, 366], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5fb40a2dbc6f7a31...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f0dcec7

Data provenance

76 FITS frames (50 MB), WASP-12251205085415.FITS → WASP-12251205123916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-251205.log (34572c2f7ee1...), FinalLightCurve_WASP-12 b_2025-12-05.png (a2bf50c5d1ed...), FinalLightCurve_WASP-12 b_2025-12-05.csv (cb3c53b33c4e...)

Compute resources

Wall time 105.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 04:47Z.